LAB ASSIGNMENT

NLP Foundations: Handwriting OCR Evaluation

Deepening your understanding after Lab 6

Course	**Artificial Intelligence**
Practical	Lab 6 of 8 - Handwriting OCR Evaluation
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Purpose

This assignment is to be completed after the Lab 6 practical. You do not need to repeat the entire experiment. Instead, use what you learned to explain the concepts, interpret the results, reason about OCR errors and suggest improvements. Answer in your own words.

Instructions

1. Complete the assignment individually.
2. Use your Lab 6 practical outputs and the results discussed during the lab as evidence.
3. Show all calculation steps. Round numerical answers to 2–3 decimal places where appropriate.
4. You may refer to your lab results; you do not need to rerun the complete OCR experiment.
5. Include screenshots or small tables only where requested. Ensure that all evidence is readable.

Part A - Understand the concepts

Q1. NLP pipeline

Describe the five stages of an NLP pipeline: acquire, normalise, represent, model and evaluate. For each stage, give an example of what happens when a handwritten medicine-name image is processed.

Five stages of an NLP pipeline:

1. **Acquire**
   A process of retrieving text from images, audio, or keyboard input.
   Example: The program loads the medicine-name into the main process.
2. **Normalise**
   Is a stage where the text is then converted to a consistent format, meaning that it first will convert every characters into lower-case form, remove spaces within the text, remove punctuations and any special characters, and only keeping letters (a-z) and numbers (0-9).
   Example:
   Initial string: Tab Azimax 40mg
   Normalised: tabazimax40mg

3. **Represent**
   A stage of converting text into a representational information that computers can process, for instance: word-counts, n-grams, and vectors.
   Example:
   - **Word-count: Tab(1) Azimax(1) 40mg(1)**
   - **N-grams: grouping adjacent words together, for example n = 2, then "Tab Azimax" and "Azimax 40mg"**
   - **Vectors:**
     - **Tab [10, 5, 20]**
     - **Azimax [20, 15, 50]**
     - **40mg [35, 25, -10]**
4. **Model**
   Application of NLP model to perform task such as classification, translation, or searching. "Tab Azimax 40mg" is classified as a medicine recipient.
5. **Evaluate**
   Comparing the system's output with the correct reference text.
   For example:
   Final result: Tab Azimax 40m9
   Correct reference: Tab Azimax 40mg
   Meaning that the system mistakenly assume g as number 9, such correction is needed for enhancing the model to learn from mistake and perform more accurately in the next analysis.

Q2. How OCR works

Explain the difference between *segmentation* and *character recognition*. Give one example of a failure at each stage. Why is it useful to identify which stage caused an error?

Segmentation: Identifying the region or places of the text located within the image, isolating texts according to the format.

Failure example: When the segmentation failed to isolate text based on the gaps, it misinterprets "to get her" into a final sentence as "together" which totally two texts with different meaning.

Character Recognition: Identifying characters within the given image and convert them into strings.

Failure example: One of the most common failure for character recognition is misinterpreting a number 1 and the letter I or l (L in lower-case) and 0 (zero) with letter O. Initial string: "S10" read as "SIO".

It is essential to understand which stage is resulting the error. Whenever the error happens to only one of the stages, an adjustment need to be conducted only to within that phase, saving development time, and reducing risk of breaking another feature.

Q3. Text normalisation

The reference is “Tab Azimax 40mg” and the OCR output is “tab azimax 40 mg”. Apply the Lab 6 normalisation rule (lowercase and keep only letters and digits). Show both normalised strings. Why must the same rule be applied to both?

	First Reference	Second Reference
Initial string	Tab Azimax 40mg	tab azimax 40 mg
Normalised	tabazimax40mg	tabazimax40mg

In order for the computers to understand the similarity and the meaning between the two texts, both have to be applied with the same rule. Without the normalization, computer will define “tab 40 mg” is not the same as “Tab 40mg” due to different space and slight upper-case.

Q4. Normalisation trade-off

The Lab 6 normaliser removes spaces. Describe one situation where removing spaces could hide an important difference. Suggest how you would change the normalisation rule for that situation.

A situation where the existence of gaps could affect the whole information in the sentence. As a result, one sentence without clear gaps generates different meanings to the sentence with clear gaps.

Suggestion: replacing every symbols and spaces with only 1 single space, separating each words highlights a clear gap and difference between normalised texts.

“Tab Azimax 40mg” ⇒ “tab azimax 40mg”

“tab azimax 40 mg” ⇒ “tab azimax 40 mg”

Part B — Work with OCR measurements

Q5. Levenshtein distance

Calculate the Levenshtein distance between “breeky” (reference) and “beeky” (prediction). Identify the edit operation(s). Then calculate the normalised edit distance by dividing the distance by the length of the longer string.

Index	Reference (breeky)	Prediction (beeky)
0	b	b
1	r	-
2	e	e
3	e	e
4	k	k
5	y	y

Operation: Deletion(‘r’)

Levenshtein Distance = 1 (as having only 1 operation)

Normalised edit distance = $\frac{LevenshteinDistance}{Max(Reference.length,\ Prediction.length)} = \frac{1}{Max(6,5)} = \frac{1}{6} = 0.16667$ **or 16.667%**

Q6. Character Error Rate (CER)

An OCR system produces 4 substitutions, 2 deletions and 1 insertion. The reference contains 35 characters. Calculate CER using CER = (S + D + I) / N. Explain what the result means. Can CER be greater than 1? Explain briefly.

N = 35

Character Error Rate (CER) = (4 + 2 + 1)/35 = 7/35 = 1/5 = 0.2 or 20%

CER measures how many characters corrections are needed compared with the reference text. In this case, meaning that 20% of the characters are corrected relative to the reference length.

CER can be greater than 1 when the total number of required edits exceeds the total length of the reference.

Q7. Compare the metrics

Explain exact-match accuracy, mean normalised edit distance, CER and blank predictions. For each metric, state whether a higher or lower value is generally better. Why is exact-match accuracy alone insufficient?

Exact Match Accuracy: Percentage of images where the normalised OCR predictions exactly matches the normalised reference text. The greater the percentage indicates the higher similarity of the images with the normalised reference text. Conversely, the lower the percentage shows the lower similarity of the images with the normalised reference text. Therefore the higher the percentage is better.

Mean Normalised Edit Distance: Average edit distance divided by the length of the longer string for every image. The more the edit distance generates higher value for the fraction, meaning that the greater the value of the MNED, there are more mistakes necessary to be corrected. Thus, the lower the fraction is indicating better process.

Character Error Rate: Total character edits divided by the total number of reference characters. Meaning that the greater the rate is, the more characters are corrected. Hence, this indicates the lower the percentage is, the better the processing.

Blank Predictions: Number of images where the normalised OCR output is empty. Empty output represents total failure of delivering the information. As a result, fewer blank predictions are better.

Exact match accuracy is insufficient due to limited diagnostic value. Without proper or detailed diagnostic value, the accuracy measurement could not differentiate whether the model suffers from characters substitution, missing texts, invalid additions, or blank outputs. Thus, several error metrics are used for overcoming this issue.

Part C — Interpret the Lab 6 experiment

Use the results from Lab 6. Both pipelines were evaluated on the same 89 labelled images, with the same normalisation rule and fixed Tesseract configuration.

Q8. Read the results

Describe at least two improvements shown by the preprocessed results. Use values from the table as evidence.

1	pipeline	exact_match_accuracy	mean_normalised_edit_distance	character_error_rate	blank_predictions
2	baseline	0.0	0.7933174191601157	0.8142857142857143	23
3	grayscale_border_resize	0.011235955056179775	0.7664388981804713	0.7857142857142857	7

- **Reduction of number of blank predictions from 23 (baseline) to 7 (preprocessed). This indicates the model has successfully outputting text from images it failed previously.**
- **Slight decrease for character error rate, falling from 81.4% (baseline) of characters error down to 78.57% (preprocessed). This demonstrates that the preprocessed images enabled the model to detect with more accurately.**
- **Slight downfall for mean normalised edit distance by going down from 79% (baseline) to 76% (preprocessed). This shows that the preprocessed images provided better readability for the model to retrieve the information.**
- **Although the exact match accuracy did not experience any significant change, the table informs a very tiny increase of matching accuracy with the gray scaled images.**

Q9. Explain the result pattern

CER and mean normalised edit distance improved, and blank predictions decreased. However, exact-match accuracy increased only from 0.000 to 0.011. Explain how preprocessing can help recover text without making the complete medicine name correct.

Preprocessed images removed noise, enhancing contrast and readability for the model by applying a grayscale filter to every image, which allows the model to recognise characters better. Although for some images that has a very high intensity of noise, such as blurs, glares, or complex font type, the processed images may only reveal partial readable characters instead of the whole full text.

Q10. Controlled experiment

Why did the lab keep the Tesseract settings (--oem 1 --psm 7), the 89 images and the normalisation rule unchanged? What would be difficult to conclude if several settings changed at once?

Maintaining these constant variables throughout the test ensures a fair baseline so that the performance improvements are affected as the result of preprocessed datasets. If multiple factors were changed simultaneously, it would make the performance shifts difficult to determine. This is

not ideal because we were targeting to prove that the gray-scaled images could essentially improve the clarity and readability for the model to read.

Q11. Diagnose the failure

Identify the likely category for each case and explain:

(a) Tesseract raises an error while processing an image;

This is a part of library runtime error when the Tesseract cannot process the input due to invalid files, corrupted files, or invalid configurations.

(b) OCR returns an empty string;

This failure shows where the OCR model runs successfully but prints out zero characters due to extreme noise, such as extreme blurs, low contrast, or any graphical disturbance within the images.

(c) OCR returns “nidu” when the reference is “nims”.

This is a part of failure in character level misclassification where the model correctly figures out the word but mistakenly assigns the correct characters as a result of similar character shapes, making it assume ‘m’ as ‘d’ and ‘s’ as ‘u’.

Part D — Apply your understanding

Q12. Suggest improvements

Suggest TWO improvements you would investigate next to improve handwritten OCR. For each, explain which error it targets and how you would test whether it helps. Do not claim an improvement works without evidence.

Two improvements that works based on personal hypothesis:

- **Adaptive cursive font reader, a feature that allows the model to inherit the capability of reading cursive or wavy texts with complex font styles, which hypothetically allowing the model to improve its quality of the predictions.**
- **Domain specific detector, a feature that allows the model to define the context before conducting any predictions, this feature hypothetically enhances the model to understand more about the context of the images. By implementing with external libraries that may introduce the model with variety of domain configuration, this will enhance the output reliability.**

Q13. Medical safety scenario

A clinic wants to use OCR to read medicine names and dosages from handwritten prescriptions. Explain at least THREE risks of using OCR output without human checking. Recommend a safer workflow.

Risks without human checking:

- **Patient harm from medication error, character substitutions error may lead the patients receiving incorrect medicine ingredients.**
- **Patients may consume overdosing or underdosing due to number recognition failure which is potentially lethal for human.**
- **Bad handwriting of high disturbance can cause blank predictions printed on patients' records, making it not reliable as a fully automated system.**

Q14. Reflection

What is the most important thing you learned from Lab 6? Identify one concept you initially found difficult and explain it using an example from the practical.

I learned a new approach on how to improve the clarity, readability, and to reduce image graphical noise is by pre-processing images before passing it to the model to retrieve the information. Even though this solution sounds simple, yet it enabled the model to perform better and managed to detect more text and characters in the given datasets.

Submission to your AI Lab Git repository

IMPORTANT: Submit your completed assignment to the Git repository you created for the AI Lab.

Do not keep the assignment only on your computer or send it only through a private message. Upload it to your existing AI Lab Git repository.

6. Create a folder named Lab06_Assignment.
7. Save your completed report as Lab06_OCR_Assignment_YourName.pdf (or .docx).
8. Put supporting screenshots or figures in a subfolder named evidence.
9. Commit and push your changes to your existing AI Lab repository(if not done yet).
10. Open the repository online and confirm that your files are visible.

Suggested repository structure

```
AI-Lab-Repository/
└── Lab06_Assignment/
    ├── Lab06_OCR_Assignment_YourName.pdf
    └── evidence/
        ├── example_1.png
        └── results_table.png
```

Assessment focus

Your work will be assessed on conceptual understanding, correct calculations, interpretation of the actual Lab 6 results, quality of reasoning, clarity of writing and completeness of the Git repository submission.

Final reminder

The purpose is understanding, not simply reproducing code or copying results. Explain what the results mean, why the experiment was designed this way and what you would investigate next.